MATH 1564 — Notes 08/27

Theorem 1: Let $Ax = b$ be a linear system on $m \times n$, and RREF of $[A \mid b]$ is $[R \mid d]$. Then:

1. $Ax = b$ has no solutions $\iff$ the d column contains a pivot.
2. If $Ax = b$ has a unique solution $\iff$ each column of R is pivotal.
3. If $Ax = b$ has infinitely many solutions $\iff$ some column of R isn't pivotal.

Proof:
1. A pivot in the d column gives the equation $0 \cdot x_1 + \cdots + 0 \cdot x_n = 1$, so no solutions.
If the d-column contains no pivot, a solution can be written down in terms of free variables, if any.

2a/2b. Suppose there is no pivot in the d column. Then the solution is unique iff there are no free variables.

Example: Given

$$[R \mid d] = \left[\begin{array}{ccccc|c} 0 & 1 & 2 & 0 & 3 & 4 \\ 0 & 0 & 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

with variables x_1, x_2, x_3, x_4, x_5. The pivots are in columns 2 and 4, so x_2, x_4 are pivotal and x_1, x_3, x_5 are free.
From row 1: $x_2 = 4 - 2x_3 - 3x_5$. From row 2: $x_4 = -1 + 2x_5$.

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} x_1 \\ 4 - 2x_3 - 3x_5 \\ x_3 \\ -1 + 2x_5 \\ x_5 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \\ 0 \\ -1 \\ 0 \end{bmatrix} + x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ -2 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} 0 \\ -3 \\ 0 \\ 2 \\ 1 \end{bmatrix}$$

Example: Given

$$[R \mid d] = \left[\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 - 2x_2 \\ x_2 \\ 2 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ 2 \end{bmatrix} + x_2 \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}$$

Interpolating polynomial:

Goal: To write a degree n polynomial $a_n x^n + \cdots + a_1 x + a_0 = p(x)$ through the data points $\begin{bmatrix} x_1 \\ y_1 \end{bmatrix}, \ldots, \begin{bmatrix} x_n \\ y_n \end{bmatrix}$.

$$\begin{cases} a_n x_1^n + \cdots + a_1 x_1 + a_0 = y_1 \\ \quad\vdots \\ a_n x_n^n + \cdots + a_1 x_n + a_0 = y_n \end{cases}$$

This is a linear system with n equations and unknowns $a_n, \ldots, a_0$.

Example: Parabola through 3 points $\begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

$p(x) = ax^2 + bx + c$

$$\begin{aligned} a(1)^2 + b(1) + c &= 2 \\ a(2)^2 + b(2) + c &= 0 \\ a(-1)^2 + b(-1) + c &= 1 \end{aligned}$$

Let the augmented matrix $= [A \mid b]$.

$$[A \mid b] = \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 4 & 2 & 1 & 0 \\ 1 & -1 & 1 & 1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & -2 & -3 & -8 \\ 0 & -2 & 0 & -1 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 0 & -2 & -3 & -8 \\ 0 & 0 & 3 & 7 \end{array}\right]$$

so $c = \frac{7}{3}$; then $-2b - 3\left(\frac{7}{3}\right) = -8 \Rightarrow b = \frac{1}{2}$; then $a + \frac{1}{2} + \frac{7}{3} = 2 \Rightarrow a = -\frac{5}{6}$.

$$\therefore \quad a = -\frac{5}{6}, \qquad b = \frac{1}{2}, \qquad c = \frac{7}{3}$$

Vectors in $\mathbb{R}^n$:

Definition: $\mathbb{R}^n$ = all matrices $\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ where each x_i is a real number.

Notation: $\mathbb{R}^1 = \mathbb{R}$.

There is a distinction between a point in $\mathbb{R}^n$, denoted $(x_1, x_2, \ldots, x_n)$, and a vector, an arrow from the origin $(0, \ldots, 0)$ to $(x_1, \ldots, x_n)$, denoted $\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$.

Informal definition: Vectors are things that can be added and scaled.

Definition: Given vectors $v_1, \ldots, v_n$ and numbers $c_1, \ldots, c_n$, the vector

$$v = c_1 v_1 + \cdots + c_n v_n$$

is a linear combination of $v_1, \ldots, v_n$ with weights $c_1, \ldots, c_n$.

Definition: The set of all linear combinations of $v_1, \ldots, v_n$ forms the span $\{v_1, \ldots, v_n\}$, the span of $v_1, \ldots, v_n$.

Example: Let $v_1 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ and $v_2 = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$. What is span$\{v_1, v_2\}$? Describe all such b.

$$c_1 \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = b$$

What can we say about b_3? b_3 is always 0.

So we actually have the system

$$c_1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}, \qquad \begin{cases} 2c_1 - c_2 = b_1 \\ c_1 + c_2 = b_2 \end{cases}$$

Is every $\begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ realized for some c_1, c_2?

$$\begin{bmatrix} 2 & -1 \\ 1 & 1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \to \begin{bmatrix} 1 & 1 \\ 0 & -3 \end{bmatrix}$$

2 pivots, yes.

Note: Since row reduction proceeds from the left, RREF of $[A \mid b]$ looks like $[\text{RREF}(A) \mid ?]$. In particular, if $\text{RREF}(A) = \begin{bmatrix} 1 & & 0 \\ & \ddots & \\ 0 & & 1 \end{bmatrix}$, then there is no pivot in the rightmost column of $\text{RREF}[A \mid b]$, so $Ax = b$ has a solution.

Thus $\text{span}\{v_1, v_2\} = x_1, x_2$-plane.

Example: If v is a nonzero vector in $\mathbb{R}^n$, then $\text{span}\{v\}$ = line in the direction of v.

Example: If $v = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$, then $\text{span}\{v\} = \left\{ \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix} \right\}$.

Goal: Write $Ax = b$ as vector equations.

Definition (matrix-vector product): Let A be an $m \times n$ matrix (m rows, n columns) with columns $a_1, \ldots, a_n$ in $\mathbb{R}^m$. Let $x = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$ be a vector in $\mathbb{R}^n$. Then

$$Ax = \begin{bmatrix} a_1 & \cdots & a_n \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = x_1 a_1 + \cdots + x_n a_n$$

= linear combination of $a_1, \ldots, a_n$ with weights $x_1, \ldots, x_n$.

Note: $\operatorname{span}\{a_1, \ldots, a_n\} = \{Ax : x \text{ in } \mathbb{R}^n\}$.

Example:

$$\begin{bmatrix} 1 & 0 & -1 \\ 0 & 2 & -2 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + 3 \begin{bmatrix} 0 \\ 2 \end{bmatrix} + 7 \begin{bmatrix} -1 \\ -2 \end{bmatrix} = \begin{bmatrix} -3 \\ -8 \end{bmatrix}$$

$(2 \times 3)(3 \times 1) = (2 \times 1)$

Example: Span of columns of $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ lies in $\mathbb{R}^2$.
Thus $Ax = b$ can be written as a vector equation $x_1 a_1 + \cdots + x_n a_n = b$.

Theorem 2: $Ax = b$ is consistent (has a solution) $\iff$ b is in the span of columns of $A = \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix}$.

Theorem 3: Let A be an $m \times n$ matrix. The following are equivalent:

1. $A_{m\times n}x = b$ consistent for every b.
2. Columns of A span $\mathbb{R}^m$.
3. A has a pivot in every row (RREF(A) has m pivots).

Proof:

$1 \iff 2$ from definition.

$3 \Rightarrow 1$: If A has a pivot in every row, then $A_{m\times n}x = b$ is consistent for every b. Wish to know RREF$[A \mid b]$ has no pivot on the rightmost column. Row reduction happens from the left, so RREF of $[A \mid b] = [\text{RREF}(A) \mid d]$ for some d. If RREF(A) has a pivot in each row, there can't be a pivot in the d-column.

$1 \Rightarrow 3$: $A_{m\times n}x = b$ is consistent for all b. Suppose the bottom row of RREF(A) has no pivot; that row is zero:

$$\text{RREF}[A \mid b] = \left[\begin{array}{ccc|c} * & \cdots & * & d \\ 0 & \cdots & 0 & d \end{array}\right]$$

Let $d = \begin{bmatrix} 0 \\ \vdots \\ 1 \end{bmatrix}$. Do the row operations taking A to R in reverse, and apply them to $[R \mid d]$; we get $[A \mid b]$ for some b that depends on d. Since $Rx = d$ is inconsistent, $Ax = b$ is inconsistent, contradiction.

Example: Is $Ax = b$ consistent for every b?

$$A = \begin{bmatrix} 1 & 2 & 0 \\ -1 & 1 & 2 \end{bmatrix} \to \begin{bmatrix} 1 & 2 & 0 \\ 0 & 3 & 2 \end{bmatrix}$$

Every row has a pivot, meaning consistent.